

Mensuration

Covering

- Units of measure
- Area and perimeter
- Circles, arcs and sectors
- Surface area and volume
- Compound shapes and parts of shapes

Units of measure

Length

- $1 \text{ km} = 1000 \text{ m}$
- $1 \text{ km}^2 = 1000000 \text{ m}^2$
- $1 \text{ m} = 100 \text{ cm}$
- $1 \text{ m}^2 = 10000 \text{ cm}^2$
- $1 \text{ cm} = 10 \text{ mm}$
- $1 \text{ cm}^2 = 100 \text{ mm}^2$

Volume

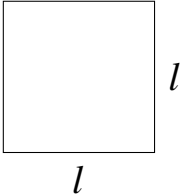
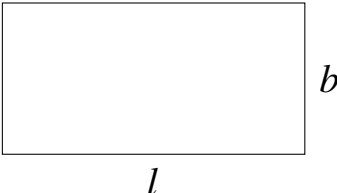
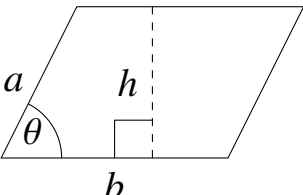
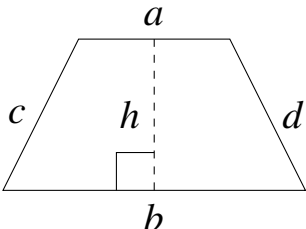
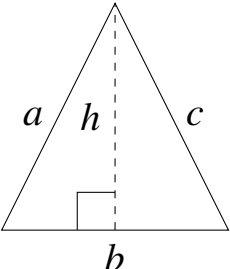
- $1 \ell = 1000 \text{ cm}^3$
- $1 \text{ m}^3 = 1000 \ell$
- $1 \text{ k}\ell = 1000 \ell$
- $1 \ell = 1000 \text{ mL}$

Mass

- $1 \text{ tone} = 1000 \text{ kg}$
- $1 \text{ kg} = 1000 \text{ g}$

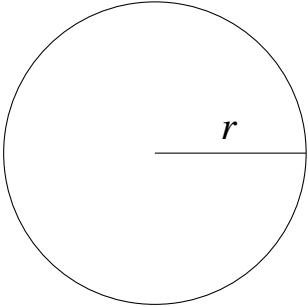
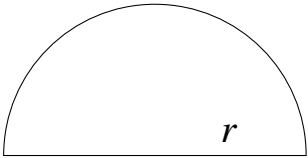
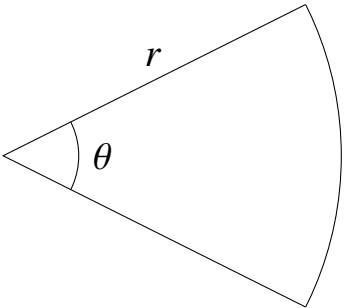


Area and Perimeter

Name	Diagram	Area	Perimeter
Square		$\text{Area} = l \times l$	$\text{Perimeter} = 4 \times l$
Rectangle		$\text{Area} = l \times b$	$\text{Perimeter} = 2(l + b)$
Parallelogram		$\text{Area} = b \times h$ or $\text{Area} = ab \sin \theta$, Where a, b sides θ is angle by sides.	$\text{Perimeter} = 2(a + b)$
Trapezium		$\text{Area} = \frac{1}{2}(a + b)h$	$\text{Perimeter} = a + b + c + d$
Triangle		$\text{Area} = \frac{1}{2} \times b \times h$ or $\text{Area} = \frac{1}{2}ab \sin C$ For C is angle by sides a and b .	$\text{Perimeter} = a + b + c$

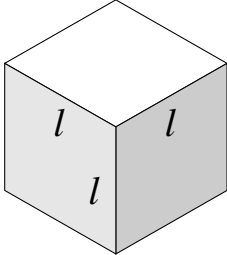
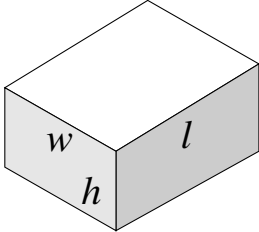
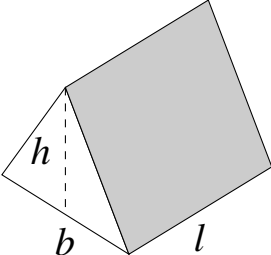
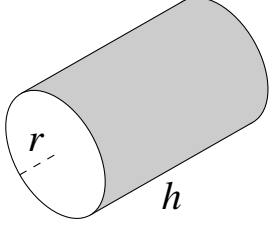


Circles, arcs and sectors

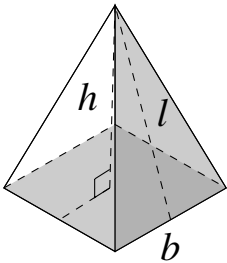
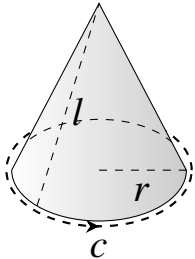
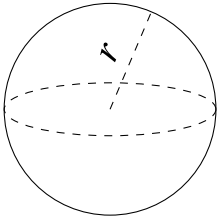
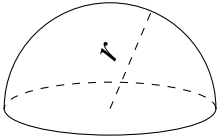
Name	Diagram	Area	Perimeter
Circle		$\text{Area} = \pi r^2$	$\text{Circumference} = 2\pi r$
Semicircle		$\text{Area} = \frac{1}{2}\pi r^2$	$\text{Perimeter} = \pi r + 2r$
Sector		$\text{Area} = \frac{\theta}{360} \times \pi r^2$	$\text{Length of Arc} = \frac{\theta}{360} \times 2\pi r$ $\text{Sector Perimeter} = \text{Arc} + 2r$



Surface area and volume

Name	Diagram	Volume	Surface Area
Cube		Volume = l^3	Surface Area = $6l^2$
Cuboid		Volume = $w \times h \times l$	Surface Area = $2(lw + lh + wh)$
Prism		Volume = base area $\times$ height For triangle prism = $\frac{1}{2}bh \times l$	Surface Area = $2 \times$ base area + (perimeter $\times$ height) For triangle prism = $bh + l(a + b + c)$
Cylinder		Volume = $\pi r^2 h$	Surface Area = $2\pi r(h + r)$ Curved Surface Area = $2\pi rh$



Name	Diagram	Volume	Surface Area
Pyramid		$\text{Volume} = \frac{1}{3} \times \text{base area} \times \text{height}$ For square pyramid $= \frac{1}{3} b^2 h$	$\begin{aligned} \text{Surface Area} &= \text{base area} \\ &+ \left(\frac{1}{2} \times l \times \text{base perimeter} \right) \end{aligned}$ For square pyramid $= b^2 + 2bl$
Cone		$\text{Volume} = \frac{1}{3} \pi r^2 h$	$\begin{aligned} \text{Surface Area} &= \pi r(r + l) \\ \text{Curved Surface Area} &= \pi r l \end{aligned}$
Sphere		$\text{Volume} = \frac{4}{3} \pi r^3$	$\begin{aligned} \text{Surface Area} &= 4\pi r^2 \end{aligned}$
Hemisphere		$\text{Volume} = \frac{2}{3} \pi r^3$	$\begin{aligned} \text{Surface Area} &= 3\pi r^2 \\ \text{Curved Area} &= 2\pi r^2 \end{aligned}$



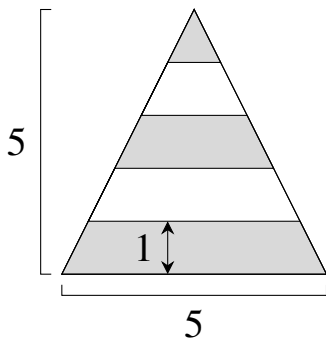
Compound shapes and parts of shapes

Compound shapes for 2d:

To find wanted area we add or subtract areas.

To find perimeter we find and add up all the outer lengths.

E.g



$$\begin{aligned}
 \square &= \text{Large Gray Triangle} - \text{White Triangle} + \text{Gray Triangle} - \text{White Triangle} + \text{Gray Triangle} \\
 A &= \frac{1}{2}(5^2 - 4^2 + 3^2 - 2^2 + 1^2) = 7.5
 \end{aligned}$$

Compound shapes for 3d:

To find wanted volumes, we add or subtract other volumes.

To find surface areas, we have to be cautious of which faces are covered or not.

E.g, The covered faces are all the circles, hence we won't count them in the final formula.

$$\text{Surface Area} = \pi r l + 2\pi r h + 2\pi r^2 = \pi(5 \times 6 + 2 \times 5 \times 10 + 2 \times 5^2) \approx 565.487$$

